Lecture 12

The measurement problem

Objectives. *An instrument gives the probabilities and conditional states of a reading. A unitary apparatus model explains those statistics, but does its global output describe one definite result? We prove a calibrated pointer cannot turn a superposition into one pointer sector or a global pointer mixture. A three-qubit test identifies what decoherence leaves open, and the premises of state completeness (C), universal unitary evolution (U) and one observer-independent global outcome (D) organise the possible responses.*

- Prove the pure-ready no-objectification theorem with complete calibration and orthogonal pointer sectors.
- Compute a global coherence test and explain the limits of moving the cut in a reading chain.
- State C/U/D and distinguish their roles from the separate assumptions of extended Wigner-friend arguments.

12.1 Calibration does not produce one global outcome

Work within one coherent sector with the full bounded-observable algebra, so that the global interference test below is admitted, rather than excluded by an additional restriction of the kind specified in L5. On $\mathcal{H}_S$ let (E_j) be a complete countable system PVM, with strong sum $\mathbb{1}$. Let (Π_j) be orthogonal pointer projections on a separable meter space $\mathcal{H}_M$, and let φ_0 be its fixed unit ready vector. A unitary U is *calibrated* if, for every unit $\psi_j \in \operatorname{ran} E_j$,

$$\chi_j = U(\psi_j \otimes \varphi_0) \in \operatorname{ran}(\mathbb{1} \otimes \Pi_j). \tag{12.1}$$

Thus every input already belonging to sector j gives the corresponding pointer reading with certainty. The output system need not remain ψ_j, and the measured sector may be degenerate. Calibration concerns all its vectors. Here calibration on an ONB already implies that by bounded linearity and the closed pointer range; calibrating only selected vectors without a spanning argument is insufficient.

Theorem 12.1 (No unitary objectification, pure ready state)**.** *Under this calibration, let the unit input ψ have at least two nonzero components $E_j\psi$. Then the global output $\chi = U(\psi \otimes \varphi_0)$ lies in no individual pointer sector. Its density operator is not any finite or countable convex mixture of density operators individually supported in pointer sectors.*

Proof. We follow each input component into its calibrated pointer range. Their superposition retains cross blocks between different ranges, whereas every mixture of definite pointer states has zero cross blocks.

Put $p_j = \left\|E_j\psi\right\|^2$, with $\sum_j p_j = 1$, and, when $p_j > 0$, $\psi_j = E_j\psi/\sqrt{p_j}$. Orthogonal strong completeness and boundedness of U give the vector-norm expansion

$$\chi = \sum_{j:p_j>0} \sqrt{p_j}\,\chi_j.$$

Calibration puts these unit vectors in orthogonal pointer ranges. Hence $\left\|(\mathbb{1}\otimes\Pi_j)\chi\right\|^2 = p_j$. At least two weights are positive, so none is one; no pointer sector contains χ. For $j \neq k$ with $p_j, p_k > 0$, the cross block is

$$(\mathbb{1}\otimes\Pi_j)\,|\chi\rangle\langle\chi|\,(\mathbb{1}\otimes\Pi_k) = \sqrt{p_j p_k}\,\left|\chi_j\right\rangle\left\langle\chi_k\right| \neq 0.$$

Every density operator supported in a single pointer sector has zero cross blocks between different sectors. Their convex mixtures retain that property, including countable mixtures: their positive omitted tail T_N obeys $\left\|(\mathbb{1}\otimes\Pi_j)T_N(\mathbb{1}\otimes\Pi_k)\right\| \leq \|T_N\| \leq \operatorname{Tr} T_N \to 0$. The displayed nonzero block therefore excludes all such mixtures. ∎

The proof uses only linearity, unitarity, a fixed product ready state and orthogonal calibration. It does not use a model of consciousness or a particular microscopic interaction. It also does not say that an input already in one sector fails to give a definite calibrated output. The obstruction is the coherent superposition of distinct sectors.

The proved statement uses a pure ready vector. Including an environment and using a mixed ready state require care about which total system the conclusion concerns.

An initially pure, separate environment can be included in the meter: replace $\mathcal{H}_M$ by $\mathcal{H}_M \otimes \mathcal{H}_E$ and include it in φ_0. Its unread records may hide all local interference, but the same global proof applies. For a mixed ready density $\tau_M = \sum_\ell s_\ell\,|m_\ell\rangle\langle m_\ell|$, suppose the mixed calibrated output for every sector input is supported in its pointer sector. Positivity then puts every $U(\psi_j \otimes m_\ell)$ with $s_\ell > 0$ in that sector. Purify the ready density by $\Phi_0 = \sum_\ell \sqrt{s_\ell}\,m_\ell \otimes f_\ell$. The extended unitary $U\otimes\mathbb{1}$ is calibrated with this pure ready vector and the enlarged pointer projections $\Pi_j \otimes \mathbb{1}$. The proved theorem therefore applies to the description including that purifying partner. Busch, Lahti, Pellonpää and Ylinen (BLPY), Theorem 22.2 p. 494, supply a different, *cited* statement about system plus probe without that partner. For a nontrivial measured observable and a nontrivial reading scale, their theorem rules out simultaneously satisfying their measurement assumptions, pointer value-definiteness, and the requirement that every initial system state yield a global system-plus-probe pointer-sector mixture. This does not say every individual output can never be pointer diagonal.

The robust version is also *cited*, Landsman, Definition 11.3 and Theorem 11.4 pp. 448–449. Let S_i be labelled sets of global density operators, with every pair from distinct sets at trace-norm distance at least $2\sqrt{1-\eta}$, $0 \leq \eta < 1/2$. For one fixed ready density τ_M and one unitary, its soundness hypothesis is explicitly

$$U(|\psi_i\rangle\langle\psi_i| \otimes \tau_M)U^\dagger \in S_i \quad \text{for every unit } \psi_i \in \operatorname{ran} E_i.$$

With this calibration into the corresponding sets and at least two nonzero input sectors, the output of some input belongs to none of the S_i. Here trace norm is the standard operator distance, normalised so density operators have distance at most two; its general construction and the robust proof are outside our standing facts. In finite dimension the trace norm of a Hermitian matrix T, with eigenvalues τ_r counted with multiplicity, is

$$\|T\|_1 = \sum_r |\tau_r|.$$

Thus computing this distance for two finite density matrices means subtracting them, diagonalising the Hermitian difference, and summing the absolute eigenvalues. This finite rule suffices for the two-state calculation on Sheet 12. The strict separation hypothesis, not merely the word "macroscopic", is part of the result.

Self-check 12.1

1. Where does the fixed ready vector enter the superposition proof?
2. Is calibration on one ONB of each sector sufficient in the pure-ready theorem? Explain.

12.2 Three premises that do not fit together

For the ideal pointer in this argument, a definite outcome j means a state supported in its pointer sector; ignorance of an already definite outcome permits a mixture of such states. That is the state–outcome identification being used. A diagonal marginal alone is not the global mixture required by it.

Definition 12.2 (The measurement trilemma)**.** The three premises in this calibrated state-and-pointer description are:

- **C**: the global quantum state is complete; there are no further actual physical facts selecting a result beyond that state.
- **U**: the same universal unitary dynamics governs the system, apparatus and any included environment.
- **D**: each run ends with one unique, observer-independent *global* outcome, represented by the specified definite pointer alternatives.

Corollary 12.3. *For the calibrated pure-ready model, an input with two nonzero spectral components, and the stated pointer-sector identification of definite outcomes, C, U and D cannot all hold.*

With C and U, Theorem 12.1 leaves the global superposed state, not a state of one of those alternatives or its ignorance mixture. D demands an additional account. The definition is a conditional organisation of these premises, not a theorem refuting every interpretation of the word "outcome". In particular, a definite outcome *relative to a branch* is not D's unique global outcome.

The operational calculus remains usable. Given a recorded result, an instrument supplies its conditional state and later probabilities. The theorem asks how a description with universal U and complete C accounts for that unique record in the first place. Renormalising a projected branch is a probability-conditioning rule; it is not by itself a unitary derivation of a stochastic physical selection.

Part V follows the three responses discussed by Maudlin (1995) and Tumulka (pp. 93–94). Bohmian mechanics adds configuration facts beyond the wavefunction, giving up C in this sense. GRW changes the unitary dynamics, giving up U. Everett retains the unitary state and takes outcomes relative to branches, giving up D's unique global outcome. These are the premises changed; their separate locality and empirical commitments must also be stated, not decided by this taxonomy. Other interpretive proposals make other specifications. We will compare the three on the same questions in L14.

A fundamental restriction of the observable algebra is another substantive specification, as L5 showed. Simply declaring pointer cross blocks unobservable would change the algebra under discussion; it is not a consequence of a small environmental overlap within the same

full algebra. Practical inability to perform a global test and a fundamental prohibition of that test are different physical premises.

Self-check 12.2

1. Which premise does each of Bohm, GRW and Everett change in this taxonomy?
2. Why is a conditional state after a recorded outcome not the missing global selection mechanism by itself?

12.3 A diagonal marginal and a global test

Take system, meter and environment qubits, in that order. Begin with $|+x\rangle \otimes |0\rangle \otimes |0\rangle$. A CNOT with system controlling meter, then a CNOT with meter controlling environment, give

$$\Psi = \frac{|000\rangle + |111\rangle}{\sqrt{2}}. \tag{12.2}$$

The pointer and system each have marginal $\mathbb{1}_2/2$. The system–meter marginal is $(|00\rangle\langle 00| + |11\rangle\langle 11|)/2$. All of them agree with restrictions of the global incoherent mixture

$$\rho_{\text{mix}} = \tfrac{1}{2}\,|000\rangle\langle 000| + \tfrac{1}{2}\,|111\rangle\langle 111|\,.$$

But the bounded self-adjoint global operator

$$B = |000\rangle\,\langle 111| + |111\rangle\,\langle 000| \tag{12.3}$$

sends Ψ to itself, so $\langle \Psi, B\Psi\rangle = 1$. Each diagonal component in ρ_{mix} has zero expectation, so $\text{Tr}(\rho_{\text{mix}}B) = 0$. The coherence has not vanished from the total state; it moved beyond the observables of the retained part. The product $\sigma_x \otimes \sigma_x \otimes \sigma_x$ gives the same two means in this example and is another possible global correlation test. The distinction between global and restricted states is shown in Figure 12.1.

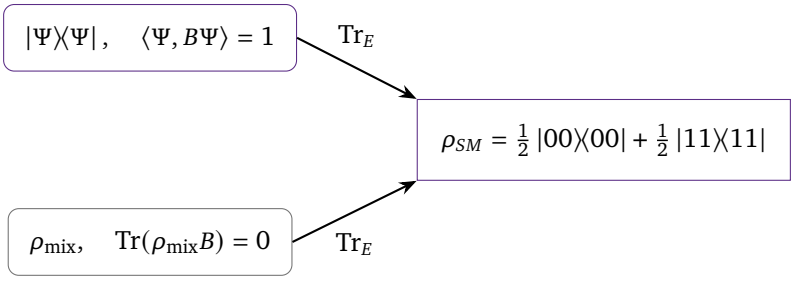


Figure 12.1: Two different global states have the same system–meter restriction. The global observable B distinguishes them, while any observable acting only on system plus meter has the same expectation.

Coherent access to every record is essential. If the environment record stays untouched and is traced out, no observable on system plus meter can recover the lost cross block. For a macroscopic environment that access may be extraordinarily difficult. Its difficulty explains the success of effective classical records, while leaving the theorem's logical distinction intact.

One can express the test as an erasure operation. Undo the two CNOTs in reverse order. The coherent state becomes $|+x, 0, 0\rangle$, whose system $+x$ probability is one. The mixture becomes $\mathbb{1}_2/2 \otimes |00\rangle\langle 00|$, whose system $+x$ probability is $1/2$. Erasing and recombining all records is a different subsequent experiment from reading and preserving their alternatives.

Self-check 12.3

1. Compute the two expectations of $\sigma_x \otimes \sigma_x \otimes \sigma_x$.
2. Which record must be coherently controlled before the inverse-CNOT test can distinguish the states?

12.4 What moving the cut preserves

A von Neumann chain includes more parts in the unitary description: system, pointer, environment, further records. In the three-qubit example, reading the meter after the first CNOT gives j with probability $1/2$ and conditional system state $|j\rangle\langle j|$. Copying that retained record to the environment gives $|jjj\rangle$. Keeping both interactions coherent and conditioning the final meter or environment record gives the same probabilities and conditional state. This is the restricted sense in which the cut can move without altering predictions.

More generally, write an instrument's record isometry (L10) as $J\psi = \sum_j K_j\psi \otimes |j\rangle$. Later controlled copying that leaves $|j\rangle$ unchanged preserves its sector weight $\left\|K_j\psi\right\|^2$ and its conditional system output. Orthogonality removes cross terms from every reading that ignores or preserves those records. Positive spectral sums give the same statement for density inputs. L14 proves this identity for the fully specified two-reading capstone.

The qualification about retained records matters. A projection update used to represent actual branch selection destroys global coherence; a coherent record-erasing unitary can detect its absence. Cut independence holds for the specified record-preserving operations. An early projection-and-mixture description and a coherent late description predict different results for a record-erasing test. Thus the claim that the cut can move always includes a restriction on what is done afterwards. The inverse-CNOT calculation explicitly distinguishes those operations. In ordinary reading chains the retained-record statistics agree; that agreement does not establish a unique global outcome.

Self-check 12.4

1. Which property of subsequent operations makes the two cut placements agree here?

12.5 Wigner's friend and the scope of further no-go claims

Take the meter in (12.2) as the friend's memory; Wigner's interference test is (12.3). Extend the chain to that memory. A unitary calibration gives a superposition of "system 0, record 0" and "system 1, record 1". Conditional on either record, the friend assigns a definite relative result. An external observer retaining coherent control of the entire isolated laboratory instead describes its global superposition and could in principle test its interference. The choice of a record reading versus a coherent global test must be included in the experimental description; they are not one unchanged experiment.

For example, conditional on memory record 0, the friend is certain that a retained system reading in the recording basis agrees with that record. Wigner is instead certain that a coherent test of B on Ψ returns +1. These certainties concern different continuations of the experiment. Transferring a prediction between agents requires specifying its observable, its conditioning record and whether that record is preserved or erased. This elementary distinction does not by itself prove any extended-friend no-go theorem.

The following named results are an optional outlook. They are not inputs to the theorem proved in this lecture. The model-relative independence conditions needed for a detailed locality comparison are defined in § 13.3.

Frauchiger–Renner (2018, arXiv:1604.07422) combine (Q) quantum probability-one inferences, (C) transfer of another agent's certainty, and (S) exclusion of contradictory single outcomes for an agent, in a scenario with coherent external control of entire labs. They argue that these premises are incompatible. Their inference-transfer condition labelled C is *not* our completeness C. This is a conditional argument about the stated inference rules and quantum-control scenario; no additional theorem is proved here and no unconditional inconsistency of quantum probability is inferred.

Bong et al. (2020, arXiv:1907.05607) formulate a different bundle called local friendliness: absoluteness of observed events, no superdeterminism of the later choices, and locality of the specified response probabilities. With suitable quantum control on the friend's scale, quantum predictions violate inequalities following from those premises. Observed-event absoluteness does not preassign every unperformed experiment; their locality premise is not Bell factorisation with outcome independence silently added. The two named arguments and this lecture's C/U/D theorem have distinct hypotheses. Listing those hypotheses is more informative than calling each one "a proof that observers disagree".

The physics

Proietti et al. (2019, arXiv:1902.05080) implemented an extended Wigner-friend scenario with a six-photon encoding of systems and records and tested a Bell-type inequality. Such photonic implementations probe controlled quantum memories under the corresponding premises. They do not demonstrate coherent manipulation of human observers or settle the interpretation of experience. The ideal global-control assumption is explicit in the mathematical scenario.

Self-check 12.5

1. After reading and preserving the friend's record, is Wigner's coherent interference test the same experimental continuation? Explain.
2. Can a unitary on system plus meter recover the interference while an orthogonal environment record remains untouched?

Does a unitary pointer end in one reading?

An instrument predicts probabilities and conditional states along a specified calibrated chain, with retained-record statistics independent of the cut. A calibrated unitary with a pure ready state does not turn a superposition into one definite pointer sector or a global sector mixture. C, U and a unique observer-independent global outcome D cannot all hold under that definite-pointer description. Decoherence explains local loss of interference but does not supply the missing selection.

Summary. The superposition proof and its nonzero global cross block establish the pure-ready obstruction. Mixed-probe and robust extensions have additional cited hypotheses. A three-qubit global test distinguishes coherence from an incoherent mixture even with identical local marginals. Moving the cut preserves the stated retained-record statistics, not arbitrary record-erasing experiments. Part V compares specified changes to C, U and D.

Exercises. Try Sheet 12 in the separate *Exercise sheets* booklet (English).

Self-check answers.

12.1 1. Every input component is tensored with the same φ_0, allowing one linear superposition under the same U. 2. Yes; bounded linearity extends from the basis span and the pointer range is closed.

12.2 1. C, U, D respectively. 2. It assumes a result has been recorded and computes its consequences.

12.3 1. One and zero. 2. The environment as well as the meter and system; an untouched orthogonal record removes local interference.

12.4 They preserve the orthogonal record labels and do not coherently recombine or erase them.

12.5 1. No. A retained record supports conditional reading statistics; the coherent test recombines alternatives and requires a different continuation. 2. No. The untouched orthogonal record removes the reduced cross block. Recovering the global interference requires coherent access to that record too.

Sources. BLPY, *Quantum Measurement*, Theorems 22.1–22.2 pp. 491–494; Landsman, *Foundations of Quantum Theory*, Theorems 11.2, 11.4 pp. 447–449; Tumulka, *Foundations of Quantum Mechanics*, pp. 93–94, 172–176 (the latter critically analyses the FR argument); Maudlin, *Topoi* **14**, 7–15 (1995). Von Neumann, *Mathematical Foundations of Quantum Mechanics*, pp. 272–273, 285, for the chain/cut discussion. Primary extended-friend papers are named in the text with their premises.